‘Microcosm’ Final Sound Design Debrief

PLEASE NOTE: The piece runs generatively for an undefined amount of time. For this reason I have chosen to include two versions of the submission as a way to not only meet the requirements of the assignment but also provide a chance for the assessor to view the piece in its entirety.

3 Minute Short Cut: https://www.youtube.com/watch?v=x2Zhxx1e-KY

Extended (as intended) Cut: https://youtu.be/LRPcRyhFEEs

For my final sound design project, I created the complete soundscape for my digital media major project, *Microcosm*, a 3D world built in Unity that explores the idea of contained human environments and the observation of life within artificial systems. The final sound design was composed and produced in **Ableton Live**, incorporating both musical and textural sound elements that respond to the visual atmosphere of the environment.

Utilising **Ableton Live’s connectivity with TouchDesigner**, I developed a reactive sound system that engages dynamically with visual changes in real time. This interactivity allowed the soundscape to evolve in sync with environmental and lighting shifts inside *Microcosm*, creating a truly multi-sensory experience.

Because *Microcosm* was inspired by micro-landscapes and the unseen world of life at its smallest scale, the challenge was to imagine what these invisible ecosystems might *sound* like. I focused on textures that felt organic yet alien, using layered synthesis, granular manipulation, and subtle low-frequency ambience to evoke an otherworldly sense of space and movement.

My workflow in Ableton centred around experimenting with **stock instruments and presets** as starting points. I would select sounds that intrigued me, tweak parameters to shape them to my vision, and then apply modulation and effects (reverb, delay, distortion, and LFO automation) to give each element depth and fluidity. The result is a mix of synthetic tones and ambient sound design that blurs the boundary between music and environmental noise.

To build melodic content without overthinking theory, I used Ableton’s **Scale feature**, which locks keys to a chosen scale. By recording improvised melodic sessions using my

laptop keyboard, I created sequences that were spontaneous yet musically coherent. These long, evolving melodies contribute to a sense of timelessness, listeners lose track of beginnings and endings as the soundscape continuously morphs.

LFOs and modulated effects added motion and unpredictability to the mix, further reinforcing the generative nature of the sound. I ensured a **clean and balanced mix** by careful EQing, gain staging, and compression, giving each frequency space to breathe while maintaining overall cohesion.

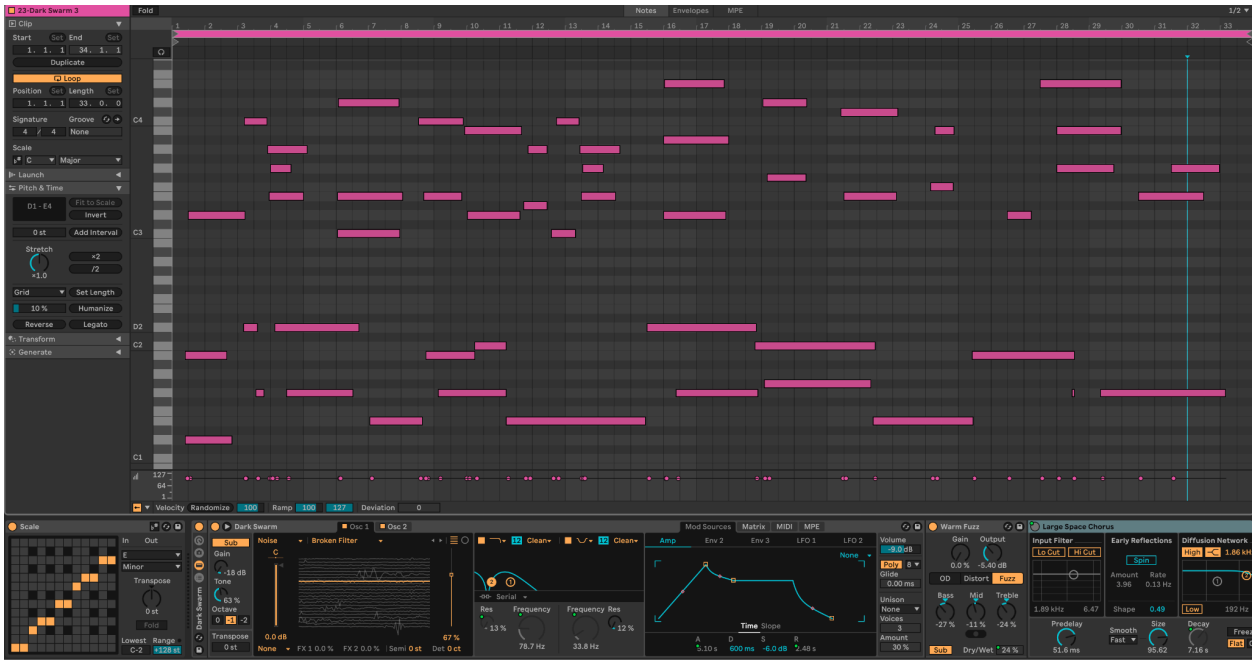
For **mastering**, I tested Ableton's built-in mastering presets to determine which enhanced the atmosphere best, then made minor manual adjustments for clarity and headroom. After initial mixing on studio monitors, I refined the final mix in the **presentation space itself** to account for the room's acoustic properties, ensuring that the immersive, low-end-driven ambience translated effectively in the gallery environment.

The final soundtrack integrates **voice, music, and environmental sound** to form a cohesive sonic narrative. It conveys a sense of isolation and observation, aligning closely with *Microcosm's* conceptual focus on human environments as experimental ecosystems.

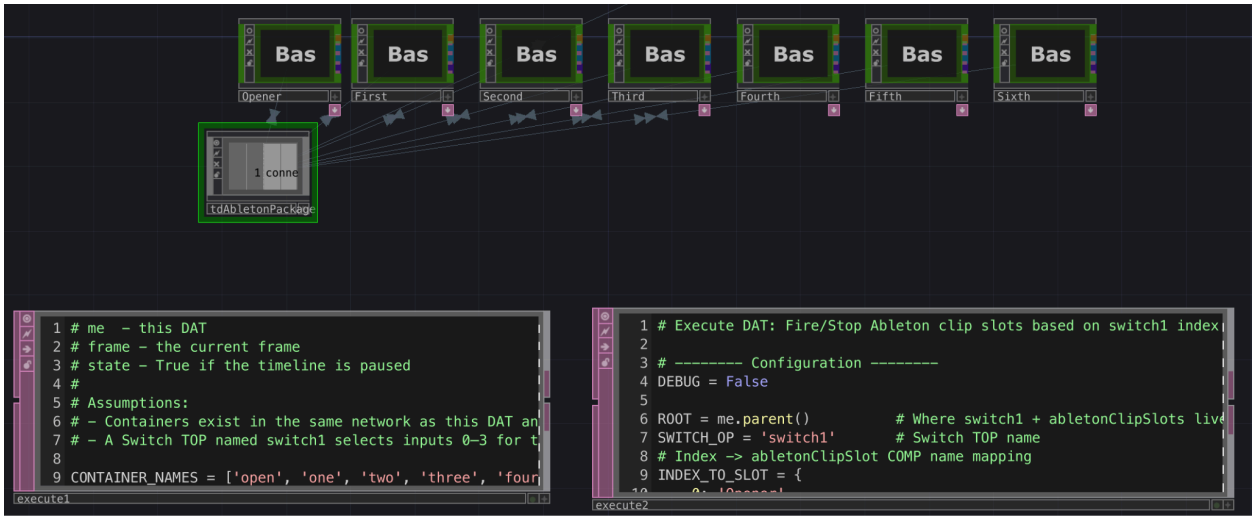
Master Chain

Use of Audio Sampling (recorded an external synth, added modulation and effects)

Utilising the plugin 'PhasePlant' for its modular synthesis abilities



Example of using Scale as a way to easily write coherent melodies. Utilising Wavetable allows for intricate texture to be applied directly into the synth



The code network for connected Ableton and TouchDesigner. Information is being sent back and forth to control parameters and fire or deactivate clips.